

SOC Monitoring & Threat Detection Lab Incident Report

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Incident Report

1. Executive Summary

This report documents a simulated brute-force login attempt detected in a SOC monitoring lab built using Splunk Enterprise, Windows Security Event Logs, and Sysmon.

Multiple failed login attempts were intentionally generated on a Windows system to simulate a brute-force attack. The events were successfully collected, indexed, and analyzed in Splunk. Detection queries, alerts, and dashboards were used to investigate the activity.

No unauthorized access occurred because the incident was performed in a controlled lab environment for educational purposes.

2. Incident Overview

Incident Type: Brute Force Login Attempt

Severity: Medium

Environment: Windows 11 Lab Machine

Detection Tool: Splunk Enterprise

Log Sources:

- Windows Security Logs
- Sysmon Operational Logs

Relevant Event IDs:

- 4624 – Successful Login
- 4625 – Failed Login
- 1 – Sysmon Process Creation
- 5 – Sysmon Process Termination

Status: Resolved (Lab Simulation)

3. Objective

The objective of this project was to design and implement a Security Operations Center (SOC) monitoring environment capable of:

- Collecting Windows Security Logs
 - Monitoring Sysmon events
 - Detecting failed authentication attempts
 - Creating security alerts
 - Building monitoring dashboards
 - Performing basic threat hunting
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4. Incident Timeline

Step	Description
1	Generated multiple failed login attempts
2	Windows recorded Event ID 4625
3	Splunk indexed the security events

Step	Description
4	Detection query identified failed logins
5	Brute Force Alert triggered
6	Dashboard updated automatically
7	Investigation completed

5. Detection Method

The following Splunk Search Processing Language (SPL) query was used to detect Windows failed authentication attempts:

```
index=* source="WinEventLog:Security" EventCode=4625
| stats count by Account_Name Source_Network_Address
```

The query performs the following tasks:

- Filters Windows Security logs for **Event ID 4625** (failed logon events).

- Groups the failed authentication attempts by **Account_Name** and **Source_Network_Address**.

- Counts the number of failed login attempts for each account.

- Helps identify potential brute-force attacks and unauthorized login attempts.

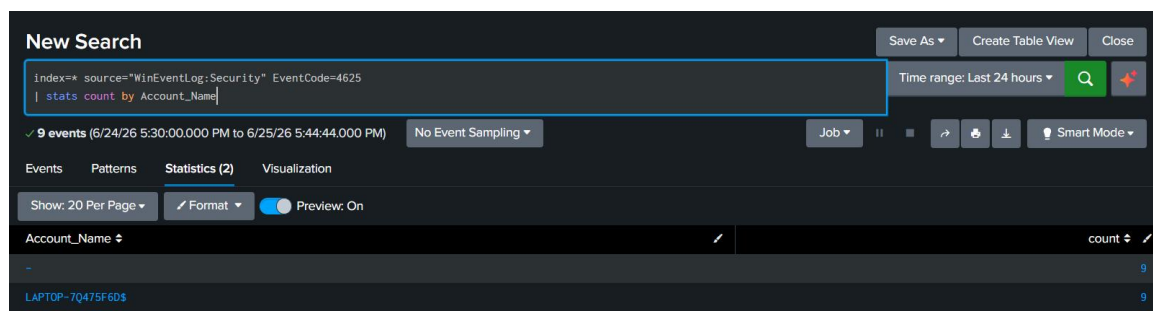


Figure 1: SPL query and output used to detect failed Windows authentication attempts (Event ID 4625).

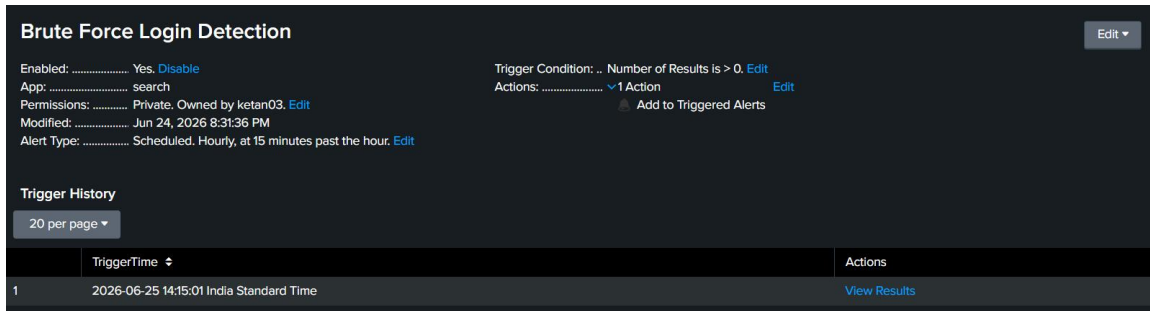


Figure 2: Scheduled Splunk alert configured to detect brute-force login attempts.

Analysis:

During testing, multiple failed login attempts were generated using invalid credentials. Splunk successfully indexed the Windows Security Event ID 4625 logs and displayed them in the search results. A scheduled alert was configured to monitor these events and notify the analyst whenever the defined threshold for failed login attempts is exceeded, enabling early detection of potential brute-force attacks.

6. Investigation

During analysis, Windows Security Event Logs were examined.

The investigation included:

- Failed login events (Event ID 4625)
- Successful login events (Event ID 4624)
- Source IP addresses associated with login attempts
- Targeted user accounts
- Sysmon process creation events (Event ID 1)
- Frequently executed processes identified from Sysmon Process Creation events (Event ID 1)

- Suspicious PowerShell and Command Prompt execution
- Process execution timeline

Analysis of the collected logs showed that the failed login attempts originated from the local machine (127.0.0.1), confirming that the activity was intentionally generated in a controlled lab environment for testing purposes.

The dashboard also confirmed successful collection of Sysmon telemetry, including process creation events and endpoint activity. Monitoring of suspicious processes such as **PowerShell** and **Command Prompt** demonstrated that the detection queries were functioning correctly.

No evidence of a successful brute-force compromise or unauthorized access was identified during the monitoring period. All detected events were intentionally generated within a controlled laboratory environment.

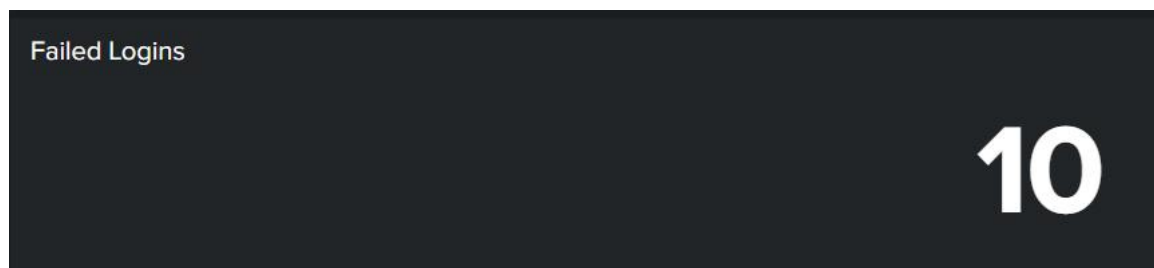


Figure 3: Dashboard panel displaying the total number of failed login attempts detected during the monitoring period.

7. Evidence Collected

The following evidence was collected during the investigation:

- Windows Security Event Logs
- Sysmon Operational Logs
- Splunk Detection Queries
- Splunk Alert Configuration

- SOC Monitoring Dashboard
- Process Creation Logs

Screenshots of the dashboard, detection queries, alert configuration, Sysmon monitoring, and investigation results are included as supporting evidence throughout this report.

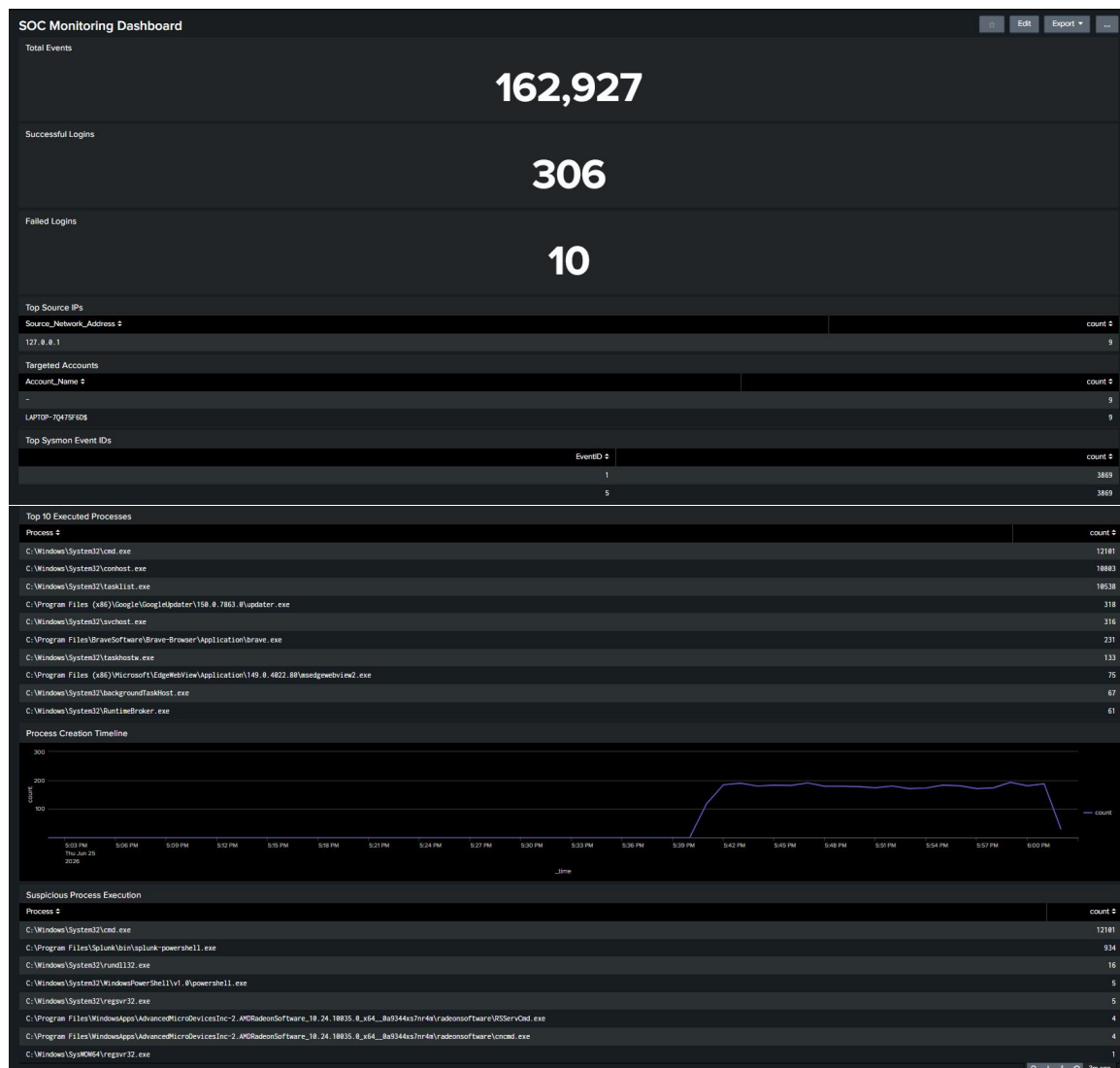


Figure 4: Complete SOC Monitoring Dashboard providing centralized visibility into authentication events, Sysmon activity, and endpoint monitoring.

Top Sysmon Event IDs	
EventID ↕	count ↕
1	4501
5	4501

Figure 5: Distribution of Sysmon Event IDs collected from the monitored Windows endpoint.

Top 10 Executed Processes	
Process ↕	count ↕
C:\Windows\System32\cmd.exe	12101
C:\Windows\System32\conhost.exe	10803
C:\Windows\System32\tasklist.exe	10538
C:\Program Files (x86)\Google\GoogleUpdater\150.0.7863.0\updater.exe	318
C:\Windows\System32\svchost.exe	316
C:\Program Files\BraveSoftware\Brave-Browser\Application\brave.exe	231
C:\Windows\System32\taskhostw.exe	133
C:\Program Files (x86)\Microsoft\EdgeWebView\Application\149.0.4022.80\msedgewebview2.exe	75
C:\Windows\System32\backgroundTaskHost.exe	67
C:\Windows\System32\RuntimeBroker.exe	61
Process Creation Timeline	
300	

Figure 6: Most frequently executed user and system processes observed through Sysmon Process Creation (Event ID 1).

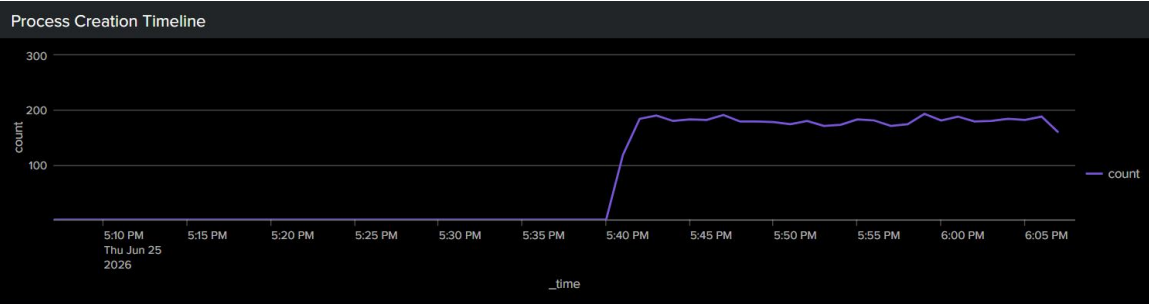


Figure 7: Timeline illustrating Sysmon process creation events throughout the monitoring period.

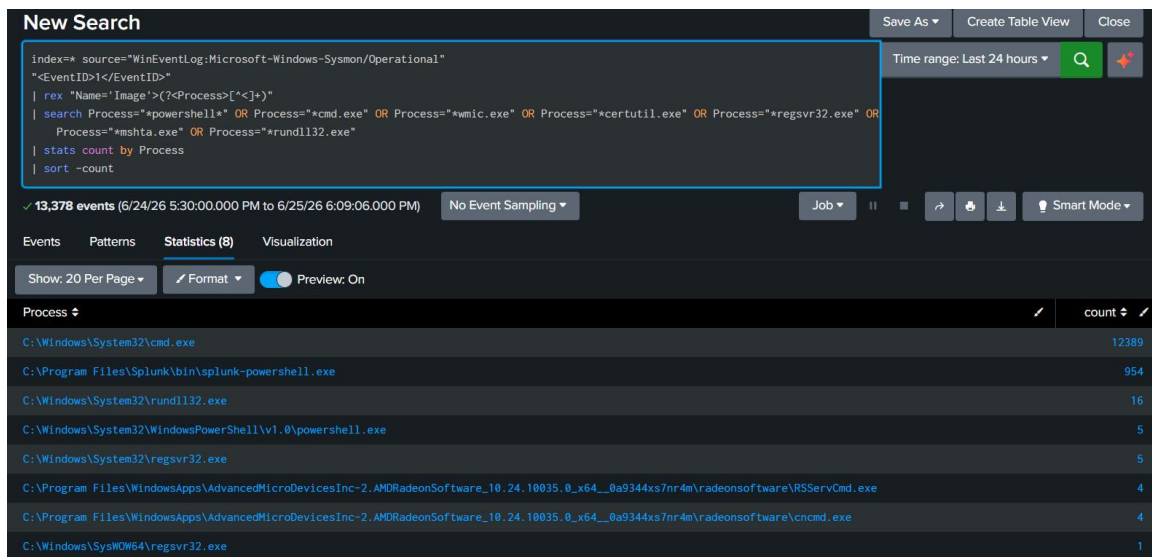


Figure 8: Detection query identifying potentially suspicious processes commonly associated with attacker techniques, including PowerShell and Command Prompt execution.

8. MITRE ATT&CK Mapping

Technique	MITRE ID
Brute Force	T1110
Valid Accounts	T1078

9. Response Actions

The following actions were performed:

- Investigated failed authentication events
- Reviewed source IP addresses
- Verified targeted user accounts

- Confirmed no successful compromise
 - Updated SOC dashboard
 - Documented findings
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10. Recommendations

To improve security, the following recommendations are suggested:

- Enable Multi-Factor Authentication (MFA)
 - Enforce strong password policies
 - Configure account lockout policies
 - Continuously monitor Windows Security Logs
 - Monitor Sysmon process creation events
 - Regularly review Splunk alerts
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11. Lessons Learned

This project provided practical experience in:

- Splunk Enterprise deployment
 - Sysmon configuration
 - Windows Event Log monitoring
 - Security log ingestion
 - SPL query development
 - Threat detection
 - Alert creation
 - SOC dashboard development
 - Basic incident investigation
 - Threat hunting techniques
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12. Conclusion

The SOC Monitoring & Threat Detection Lab successfully demonstrated how Windows Security Logs and Sysmon can be integrated with Splunk Enterprise to detect authentication-related threats and monitor endpoint activity.

The project simulated a brute-force login attack, generated relevant security events, created detection queries and alerts, and visualized security data through an interactive dashboard.

This project demonstrates practical SOC skills including Windows log analysis, Sysmon monitoring, SIEM configuration, SPL query development, security alert creation, dashboard visualization, and basic incident response. It serves as a hands-on cybersecurity portfolio project showcasing real-world Security Operations Center workflows.

13. References

- 1) Splunk Enterprise Documentation
- 2) Microsoft Windows Security Auditing Documentation
- 3) Microsoft Sysmon Documentation
- 4) MITRE ATT&CK Framework
- 5) Sysinternals Sysmon